

Day 2 Task

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import cv2
import mediapipe as mp
mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils
mp_drawing_styles = mp.solutions.drawing_styles

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=1,
    smooth_landmarks=True,
    enable_segmentation=False,
    min_detection_confidence=0.5,
    min_tracking_confidence=0.5
)
cap = cv2.VideoCapture(0)
if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()
print("Pose Detection Running... Press 'Q' to quit.")
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        print("Failed to grab frame.")
        break
    frame = cv2.flip(frame, 1)
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    rgb_frame.flags.writeable = False
    results = pose.process(rgb_frame)
    rgb_frame.flags.writeable = True
    frame = cv2.cvtColor(rgb_frame, cv2.COLOR_RGB2BGR)
    if results.pose_landmarks:
        mp_drawing.draw_landmarks(
            frame,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS,
            landmark_drawing_spec=mp_drawing_styles.get_default_pose_landmarks_style()
        )
    landmarks = results.pose_landmarks.landmark
    h, w, _ = frame.shape

    key_points = {
        "L.Shoulder": mp_pose.PoseLandmark.LEFT_SHOULDER,
        "R.Shoulder": mp_pose.PoseLandmark.RIGHT_SHOULDER,
        "L.Elbow": mp_pose.PoseLandmark.LEFT_ELBOW,
        "R.Elbow": mp_pose.PoseLandmark.RIGHT_ELBOW,
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        "L.Hip": mp_pose.PoseLandmark.LEFT_HIP,
        "R.Hip": mp_pose.PoseLandmark.RIGHT_HIP,
        "L.Knee": mp_pose.PoseLandmark.LEFT_KNEE,
        "R.Knee": mp_pose.PoseLandmark.RIGHT_KNEE,
    }

    for label, point in key_points.items():
        lm = landmarks[point]
        cx, cy = int(lm.x * w), int(lm.y * h)
        cv2.putText(frame, label, (cx + 8, cy - 8),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.45, (0, 255, 255), 1, cv2.LINE_AA)

    status = "Pose Detected"
    color = (0, 255, 0)
else:
    status = "No Pose Detected"
    color = (0, 0, 255)
cv2.rectangle(frame, (0, 0), (frame.shape[1], 40), (30, 30, 30), -1)
cv2.putText(frame, f"MediaPipe Pose Detection | {status}", (10, 27),
            cv2.FONT_HERSHEY_SIMPLEX, 0.65, color, 2, cv2.LINE_AA)
cv2.putText(frame, "Press Q to quit", (10, frame.shape[0] - 10),
            cv2.FONT_HERSHEY_SIMPLEX, 0.5, (180, 180, 180), 1, cv2.LINE_AA)

cv2.imshow("Pose Detection - AI/ML Intern Demo", frame)

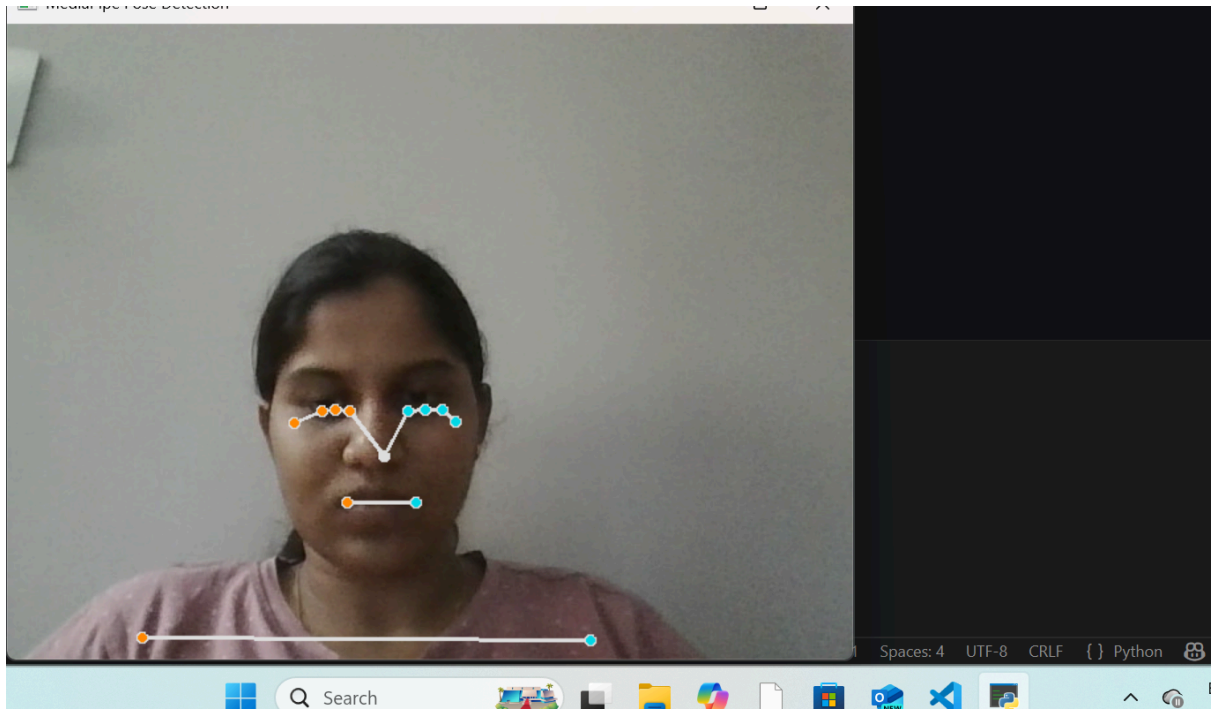
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
pose.close()
print("Closed.")
import cv2
import mediapipe as mp

# Initialize MediaPipe Pose
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    min_tracking_confidence=0.5
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```



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